Day 6: Introduction to Network Devices

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Topic: Day 6: Introduction to Network Devices

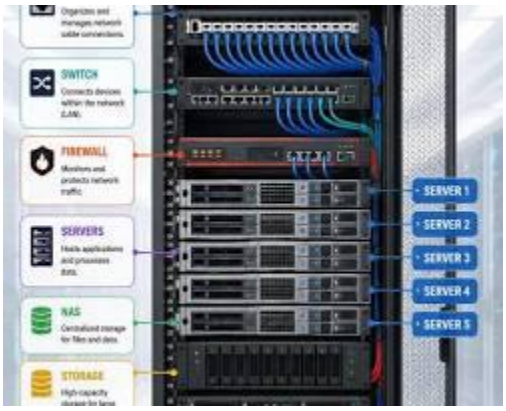
Introduction

Network devices help computers and other devices communicate. A switch connects devices inside a local network, a router connects different networks, an access point provides wireless access, a firewall controls traffic, and a hub repeats traffic to its ports.

Task 1

Device	Main Function	Typical Ports/Interfaces
Switch	Connects devices in a LAN	Ethernet, console, power
Router	Connects different networks	Ethernet/WAN, console on some models, power
Hub	Repeats traffic to all ports	Ethernet, power
Firewall	Allows or blocks network traffic	Ethernet/WAN/LAN, management interface
Access Point	Provides wireless network access	Ethernet, power, antennas on some models

Task 2



Use pictures or diagrams of at least three devices and label their ports. Suggested labels:

• Switch: Ethernet ports, console port, power.

• Router: Ethernet/WAN ports, console if present, power.

• Access Point: Ethernet port, power, antennas if the model has external antennas.

Task 3

If I use Instagram on my phone, the access point provides the Wi-Fi connection between my phone and the local network. The router sends traffic between my local network and the Internet. In a larger network, switches and firewalls can also be involved before traffic reaches external services.

Task 4

Device	**Possible issue**	**How an upgrade may help**
Access Point	Too many users or weak coverage	Newer model can support more users and better coverage
Switch	Too much local traffic or limited capacity	A suitable higher-capacity switch can handle more traffic
Router	High traffic or limited performance	A newer router can provide better processing and wireless/network capacity

Conclusion

Different network devices have different jobs. Knowing these jobs makes it easier to understand a network and find the likely cause of a problem.

Quick Revision

- Switch = connects devices in a LAN.
- Router = connects different networks.
- Access point = provides Wi-Fi.
- Firewall = controls allowed and blocked traffic.
- Hub = repeats traffic to all ports.